

(K, Na)NbO₃-Based Lead-Free Piezoceramics: Fundamental Aspects, Processing Technologies, and Remaining Challenges

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Environment-friendly lead-free piezoelectric ceramics have been studied extensively in the past decade with great progress particularly in systems based on a niobate perovskite compound formulated as (K, Na)NbO₃ (abbreviated as KNN). A comprehensive review on the latest development of KNN-based piezoelectric ceramics is presented in this article, including the phase structure, property enhancement approaches, and sintering processes as well as the status of some promising applications. The phase structure of KNN was reexamined and associated with the effect of chemical modification on its tetragonal-to-orthorhombic transition. Then, a special focus is placed on the temperature dependence of piezoelectric properties of KNN-based ceramics, followed by reviewing the recent approaches devoted to the temperature-stability enhancement. The processing fundamentals related to the sintering of KNN-based ceramics are also presented with an emphasis on compositional and microstructural control. Finally, this review introduces several industrial attempts of traditional piezoceramic products using KNN-based ceramics and the studies on some promising application in authors' laboratory.

I. Introduction

PIEZOELECTRIC materials have the functionality of producing an electrical potential in response to an applied force or generating mechanical movement when subjected to an electric field. Such smart materials have become essential for modern society, especially in the fields of information and communications, industrial automation, medical diagnostics, etc. Typical applications include actuators, sensors, transducers, so on. Although tremendous innovations are now available with piezoelectric materials, the terminology of piezoelectricity is not new. The appearance of an electric polarization in quartz single crystals that were subjected to a mechanical load was first discovered by the Curie brothers in 1880,¹ whereas the effect that links a mechanical action

(stress or strain) with an electric response (electric field, displacement, or polarization) is named the piezoelectric effect or, more precisely, the direct piezoelectric effect. Later, it was experimentally confirmed that the converse piezoelectric effect is also observed in single crystals with acentric symmetry in that an external electric field generates a mechanical response, i.e., a stress or strain of the sample, similar to the electrostriction of dielectrics. It is generally recognized that the intrinsic piezoelectric effect reflects a linear relationship between electrical and mechanical variables and originates from the displacement of ions of a single crystal with acentric symmetry under an applied electric field. This relationship enables a change in the sign of the effect by switching the external electric field, unlike the electrostriction that always exhibits a quadratic expansion.²

Although piezoelectricity is found in several types of natural materials, most modern devices use polycrystalline ceramics, which can be easily fabricated and shaped at low cost. For decades, lead zirconate titanate (PZT) ceramics have been market dominating due to its excellent properties and flexibility in terms of compositional modifications. However, the large amount of lead contained in PZT materials has drawn much attention during the past decade due to the environmental concern as well as governmental regulations against hazardous substances. In 2006, the European Union has already adopted the well-known directives waste electrical and electronic equipment (WEEE) and Restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS),^{3,4} with the purpose of protecting human health as well as environment by exclusion or substitution of hazardous substances used in electrical and electronic devices. Similar regulations are also planned or established in North America, Japan, Korea, and China. Lead ranks foremost among the hazardous substance list due to its high toxicity with maximum allowed concentration of merely 0.1 wt%. However, PZT ceramics, which contain more than 60 wt% lead oxides, are still exempted in piezoelectric devices up to now. Therefore, tremendous efforts have been devoted to the development of competitive lead-free counterparts, such as BaTiO₃, Bi_{1/2}Na_{1/2}TiO₃, (K, Na)NbO₃ (KNN), and their derivatives. At the moment, great embarrassment has been encountered in search for one single qualified lead-free system substituting for PZT that is so versatile in properties by utilization of various doping elements

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that it can fit almost into all purposes. In our opinion, this is not a problem as it would be easier to exploit the outmost potential of various lead-free systems to substitute for different types of PZT.

Among all the lead-free candidates, alkali niobate ceramics based on KNN have drawn much attention since Saito *et al.* made the breakthrough in Li-, Ta-, and Sb-modified KNN ceramics with textured structure in 2004.⁵ Although pure KNN ceramics is normally considered to have a moderate piezoelectric constant d_{33} around 80 pC/N,⁶ the value reported by Saito *et al.* reached amazingly up to 416 pC/N, comparable to that of soft PZT ceramics. On the other hand, a significant feature of KNN ceramics lies in the inherent compatibility with nickel electrode,^{7,8} which is absent in the PZT case but very important for industry due to the significant reduction in processing cost. However, we are fully aware of the fact that new materials are not likely to succeed in application unless it offers far more advantages than its conventional counterparts. Technically, lead-free piezoceramics, including the KNN-based ones, are able to compete with existing PZT families in certain applications; however, it is hard for them to surpass PZT with substantial benefit. Therefore, it is said that a successful transfer of lead-free piezoceramics technology to industries is critically dependent on the strength of governmental legislation.

It is noted that the present work manifests itself by undivided dedication to the KNN-based system, while there are several excellent reviews concerning comparative studies of various lead-free piezoceramics.^{9–12} As shown in Fig. 1, hundreds of publications concerning the development of KNN ceramics have been introduced over the last decades, indicating indisputable interests on this system as the most promising candidate for commercially viable lead-free piezoceramics candidate. In the present article, we will provide insightful assessment on current status of KNN-based ceramics, based on a comprehensive review of available publications of significance. It is our great pleasure if this work could provide useful references and shed light on the development of KNN-based materials in the future.

II. Phase Structure

It is surprising that $K_xNa_{1-x}NbO_3$ ceramics were investigated intensively several decades ago both in material properties and structures,^{13,14} as it was found as a piezoelectric

material in as early as 1950s.⁶ Although $KNbO_3$ and $NaNbO_3$ can form solid solutions with a perovskite structure regardless of x value, as shown in Fig. 2, they have distinct symmetries. Generally, pure $KNbO_3$ has the following polymorphisms, low-temperature (less than -10°C) rhombohedral (R) phase, room-temperature orthorhombic (O) phase, and high-temperature tetragonal (T, $225^\circ\text{C} \sim 435^\circ\text{C}$) and cubic (C, $>435^\circ\text{C}$) phase.¹⁵ In contrast, phase transitions of $NaNbO_3$ are more complicated that there are as many as seven polymorphs across the whole temperature range.¹⁶ The room-temperature polymorph of $NaNbO_3$ was reported to be orthorhombic and antiferroelectric, with the space group $Pbma$. The complexity of the structure of $NaNbO_3$ is strengthened by an oxygen octahedral tilting, which is absent in $KNbO_3$. This could be possibly due to the fact that K is relatively large enough to fill the 12-coordinated cavity of the ideal perovskite structure. It is interesting to note that only 2% of K is sufficient to change the initial structure of $NaNbO_3$ toward that of $KNbO_3$; however, octahedral tilting exists for a wide composition range until K/Na ratio approximately equal to 1:1,¹⁴ around which the piezoelectric performance peaks in the solid solutions of $K_xNa_{1-x}NbO_3$.^{6,17–19} Hence, we intend to refer to the compositions around $(K_{0.5}Na_{0.5})NbO_3$ for the KNN ceramics mentioned hereafter if no specific explanation is provided. Certain reports including some early ones considered that there are two different orthorhombic phases (O_1 and O_2) around $(K_{0.5}Na_{0.5})NbO_3$ forming a morphotropic phase boundary (MPB), although the exact K/Na ratio varies slightly depending on the reports,^{13,18} as shown in Fig. 2. Although the presence of this MPB is still controversial in the literature yet,^{20,21} it is undeniable that the dielectric and piezoelectric properties of KNN are maximized in the vicinity of $(K_{0.5}Na_{0.5})NbO_3$ composition even in the presence of dopants.^{19,21–24}

With increasing temperatures, KNN shows several phase transitions similar to $KNbO_3$,²⁵ as depicted in Fig. 3. The rhombohedral polymorph is not mentioned here as it only appears at very low temperatures (~ 100 K) and in some cases it is not easy to be identified.^{20,26} KNN is known to have an orthorhombic structure with the space group $Amm2$ at room temperature; however, the situation has a level of complexity in that the structure can be seen in two different ways, namely, strictly speaking, the proper designation in the context of crystallography should be an orthorhombic structure with a cell doubling, although the perovskite-type ABO_3 primary cell possesses monoclinic symmetry.¹³ As a result,

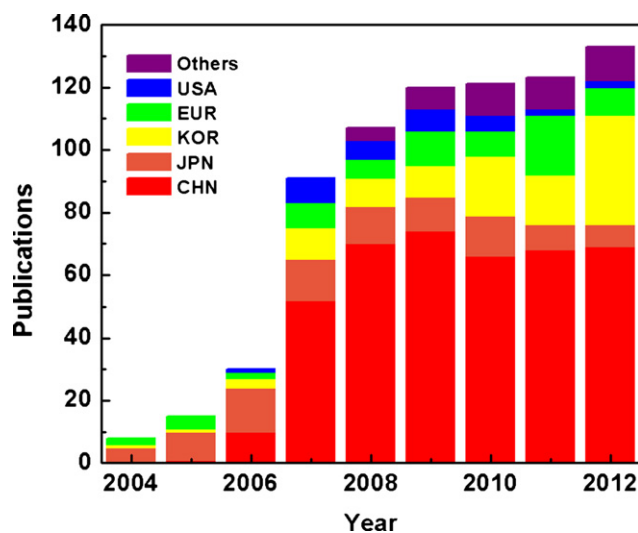


Fig. 1. The number of publications on KNN-based lead-free piezoceramics from 2004 to 2012. The search was conducted in “ISI Web of Science” by counting the relevant articles with the keywords of “lead-free”, “piezoceramics”, and “(K, Na)NbO₃”.

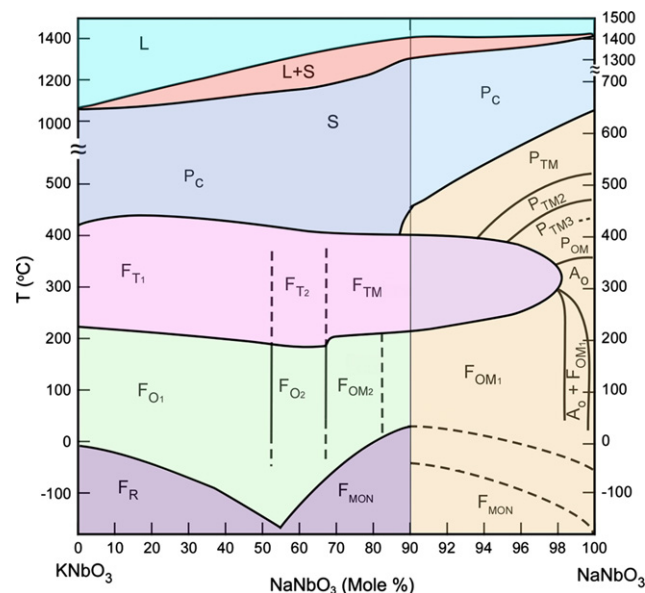


Fig. 2. Phase diagram of $KNbO_3$ – $NaNbO_3$ pseudobinary solid solutions.¹

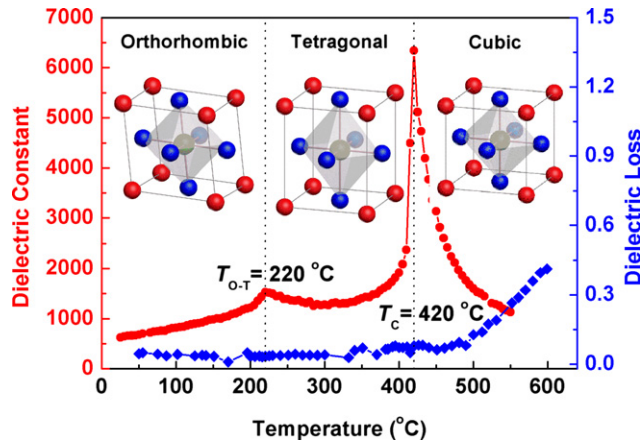


Fig. 3. Temperature-dependent dielectric permittivity and loss of $K_{0.5}Na_{0.5}NbO_3$ ceramics, with schematic plots of various polymorphs. The red, green, and blue balls represent alkali elements in A-sites, Nb atoms in B-sites, and oxygen atoms, respectively.

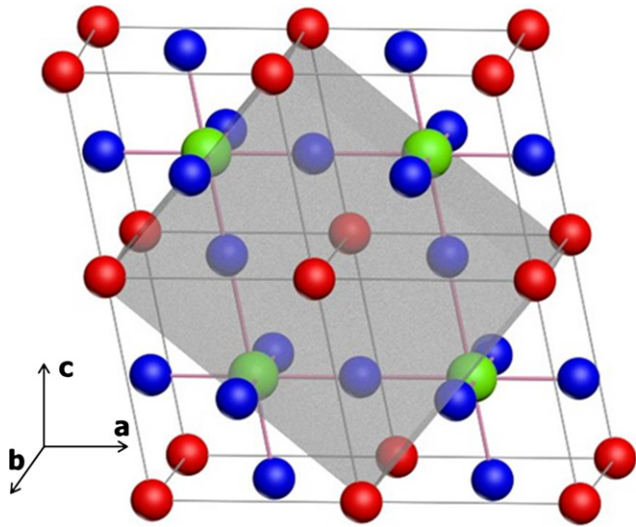


Fig. 4. Schematic illustration shows a consistency between orthorhombic symmetry and monoclinic ABO_3 primary cell. The red, green, and blue balls represent A-sites, B-sites, and oxygen atoms, respectively.

crystallographic indexing is often inconsistent in literatures, especially for researchers mainly focusing on materials properties. It should be noted that the monoclinic primary cell is a special case where two axes equal to each other, viz., $a_m = c_m$; while b_m ($a_m = c_m > b_m$) axis is perpendicular to $a_m c_m$ plane and its interaxial angle β is little more than 90° .²⁷ Tellier *et al.* pointed out that there is a small difference between a_m and c_m in length (~ 0.002 Å), and, thus, assigned the monoclinic symmetry for the whole matrix.²⁰ Indeed, the difference is so small that the peak splitting cannot be resolved with the use of normal X-ray diffraction (XRD) techniques. As this review put more weight on the practical aspect of the materials of concern, we refer to the orthorhombic symmetry throughout the manuscript.^{25,26,28–30}

Four adjacent monoclinic ABO_3 primary cells are schematically presented in Fig. 4. It is noticed that the deviation of angle β from 90° is exaggerated for clearer presentation of the structure. As a_m axis equals c_m axis in length, the diagonals (gray area) in Fig. 4 form a rectangle in $a_m c_m$ plane, which is the projection of unit cell of KNN along b_m axis. Hence, it could be easily understood that the perovskite-type primary cell of KNN is monoclinic while its unit cell has an orthorhombic symmetry at room temperature.

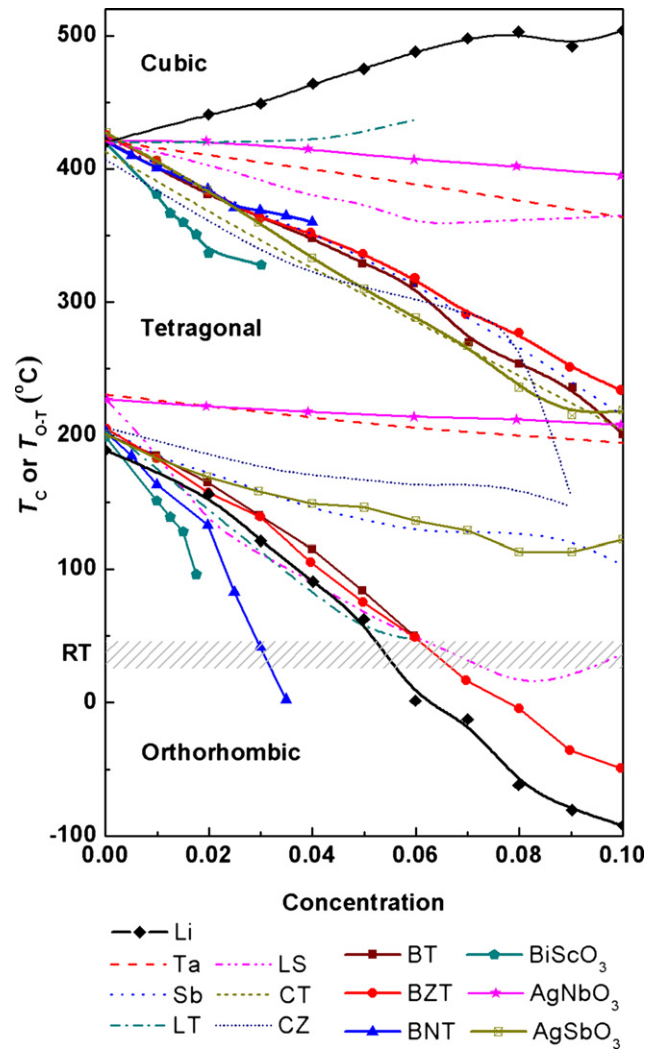


Fig. 5. Effect of different substitutions on the phase transition temperatures of KNN ceramics.^{30,48–59}

When temperature exceeds 220°C , a phase transition from orthorhombic to tetragonal occurs in KNN (Fig. 3) with the space group changing from $Am\bar{m}2$ to $P4mm$,³¹ where the cell doubling disappears with the perovskite-type ABO_3 primary cell being the unit cell itself. The phase structure becomes cubic at temperatures above the Curie temperature T_C around 420°C .

The phase transition temperatures in KNN system can be easily shifted by chemical doping, as depicted in Fig. 5. Typical A-site dopants are Li, Ag, Ba, Ca, whereas Ta, Sb, Zr, etc. are normally considered to enter the B site. Among them, Li doping has been extensively investigated as a representative modifier for this system,^{32–42} resulting in downward shift of the orthorhombic–tetragonal phase transition temperature (T_{O-T}). The movement of T_{O-T} toward room temperature favors the enhancement of piezoelectricity, known as polymorphic phase transition (PPT) effect.^{43,44} Both PPT and typical MPB in PZT system allow the performance to be largely enhanced with a two-phase coexistence. The presence of PPT in KNN-based materials is usually characterized by XRD using peak splitting of (222) peaks and intensity variations of (002)/(200), or more preferably (004)/(400) peaks.²⁷

On the other hand, an additional triclinic symmetry has been reported for the special case of very small particles of about 130 nm,⁴⁵ which is probably due to the cationic ordering of potassium and sodium in the unit cell, induced by the very small number of cells in the particles. Except for normal phase transition detecting techniques such as XRD and

Raman spectroscopy, an extended X-ray absorption fine-structure analysis was used to determine neighborhood of Nb atom in KNN system, revealing independent Nb displacement with the symmetry of the crystal lattice, which is counterintuitive and may need further investigation.^{46,47}

III. Property Enhancement

In general, the piezoelectric output in ferroelectric ceramics is composed of intrinsic and extrinsic contributions. The former refers to linear piezoelectric effect of lattice displacement, whereas the latter originates mainly from movement of domain walls.⁶⁰ As derived from the Landau–Devonshire–Ginzburg phenomenological relation, the linear piezoelectric effect is proportional to the magnitude of P_s ; therefore, the larger P_s , the higher the piezoelectric coefficient is expected. Typical P_s values of KNN-based materials are in the range from 15 to 25 $\mu\text{C}/\text{cm}^2$, comparable to those of PZT types. One may suggest that the discrepancy in the piezoelectric properties between KNN and PZT could be largely due to the difference in the extrinsic contribution. However, it is considered by the authors that clear distinguishment between these two contributions may be impractical. For example, the enhancement in dielectric, piezoelectric, and electromechanical coupling properties along the MPB in PZT-based ceramics is normally attributed to facilitated polarization switching, which holds true even for monodomain single crystals, indicating the intrinsic nature of MPB.^{61,62} However, it is, of course, undeniable that facilitated domain wall movement in the MPB region would further benefit the piezoelectric performances.⁶³ In any case, we can conclude that the control over the phase structure is important for piezoceramics; particularly creating a two-phase coexisting state is preferred, which can be either due to compositionally driven MPB or temperature-driven PPT effect. The manipulation of phase structures in KNN is usually realized via compositional modification, which is the first topic in this part. In addition, alternative is a domain engineering, which can enhance piezoelectric performances in KNN-based ceramics. In the last part of the present section, we review the issues concerning temperature instability arising from the PPT effect and discuss possible solutions.

(1) Chemical Modification

As aforementioned, KNN ceramics by conventional sintering is reported with a moderate piezoelectric constant d_{33} around 80 pC/N half a century ago,⁶ which was improved to ~125 pC/N recently.^{21,64–66} If the K/Na ratio is elaborately optimized, higher d_{33} even up to 160 pC/N can also be achieved.¹⁸ It should be noted that the d_{33} could be improved to 150–160 pC/N using pressure-assisted techniques such as hot pressing (HP) and spark plasma sintering (SPS), without intentional optimization of K/Na ratio.^{17,67–72} Not many reports concerning chemical modification in this system come out until the year 2004,^{73–75} when a breakthrough was made by Saito *et al.* that a significantly high d_{33} over 400 pC/N was realized in KNN by multiple dopants of Li, Ta, and Sb, as well as well-controlled texturing technique.⁵ Actually, it is considered that this discovery has opened up a seeming campaign in academia for lead-free piezoceramics, as reflected in Fig. 1. At first, it was believed that dopants like Li, Ta, and Sb could consist in the constitution of MPB in KNN-based ceramics, which mimicked that in PZT ceramics and eventually resulted in the property enhancement. Thus, all kinds of dopants have been exploited with a special emphasis on the improvement of piezoelectric constant d_{33} , which is usually measured using the convenient quasi-static method (Berlin-court meter method) and taken as the small signal d_{33} . Typical d_{33} values obtained are among 200–300 pC/N. Later on, it was proposed that the underlying mechanism for property enhancement of KNN-based ceramics by means of these

manners is not the classical MPB but PPT effect,^{43,76} which exists in many materials such as BaTiO₃. The PPT theory claims that property enhancement in KNN-based ceramics is due to tetragonal–orthorhombic PPT point (T_{O-T}) shifting downward from around 200°C toward room temperature, as shown in Fig. 5 as well as Tables I and II. However, MPB is actually more favorable over PPT as the enhanced properties from PPT, in definition, should suffer from temperature instability, which will be discussed later. As well, it should be noted that texturing is quite effective for property enhancement, although very challenging for KNN-based ceramics.^{5,77} In addition, the grain size effect is considered important for structures and properties in piezoceramics; however, discrepancy exists for KNN-based materials.^{78,79}

Although all the dopants shown in Fig. 5 shift T_{O-T} to room temperature, the obtained d_{33} data vary depending on the type of dopants. In other words, the apparent orthorhombic–tetragonal coexistence manifested by XRD pattern does not guarantee high d_{33} , e.g., 300 pC/N or even above. In fact, the most commonly obtained values are among 200 ~ 250 pC/N, which could be attributed to several reasons. One of the most important reasons considered by the current authors is that the ratio between the two phases might be important, which means that if the T_{O-T} is decreased quickly, the optimal phase constitution is difficult to be obtained at room temperature.¹⁰³ This consideration is manifested in the case of Li doping, where the discrepancy in d_{33} among various reports can be as large as around ± 150 pC/N.^{32,33,35,37–39,81} The second one is that the phase structure and properties of KNN-based ceramics are so sensitive to processing conditions, such as sintering temperatures, that sometimes the reproducibility is not guaranteed, which will be discussed later. As well, a complex stress situation in polycrystalline materials can play a significant role in the final properties as reported that the T_{O-T} can be influenced as much as tens of degrees even due to intergranular stress.¹¹⁸

A significant feature that should be pointed out is that among all these dopants and species, only Li and some Li-containing species can increase the Curie temperature T_C , as shown in Fig. 5. With a solubility limit of around 10 mol %, the highest T_C value can reach up to 500°C, which makes a promising candidate for high-temperature applications. Except for Li, the T_C of KNN-based ceramics is generally lowered with the introduction of chemical modifiers, although some of them, such as Ag, show a relatively mild effect. One should bear in mind that the trend shown in Fig. 5 is the consequence of doping in pure KNN ceramics, and it is possible to combine several kinds of dopants, resulting in more complex outcome. However, certain common regularities can be drawn out of multidoping situations, as shown in Fig. 6, which shows the relationship between T_C and d_{33} . It should be noted that the compositions here are chosen with the criteria that the materials maintain orthorhombic–tetragonal two-phase coexistence at room temperature. The trend is generally in accordance with that in PZT ceramics,⁷⁶ which means that higher T_C is always combined with lower d_{33} and *vice versa*. It is indicated in Fig. 6 that the achievement of high d_{33} normally requires compositional complexity and that improved piezoelectric performances in monodoped KNN requires special treatment, such as aging and repoling treatment as indicated in Fig. 6,⁴⁰ which will be discussed in the next part of this section.

As T_C is actually influenced by the c/a ratio in tetragonal perovskites, Fig. 7 is presented for a better understanding. As the data here represent the compositions with orthorhombic–tetragonal coexisting phases, the c/a ratios, which are mostly calculated from peak intensities of (002)/(200), indeed reflects distortions apart from ideal cubic ABO₃ cell. On the other hand, the relationship between c/a and d_{33} is rationalized in that large c/a implies large spontaneous polarization but not necessarily high d_{33} , as smaller c/a favors domain wall motions due to reduced internal compatibility stress and

Table I. Properties of KNN and Li, Ta, and Sb-KNN Piezoceramics

Material	d_{33} (pC/N)	k_p (%)	ϵ_r	$\tan \delta$	Q_m	T_C (°C)	T_{O-T} (°C)	c/a	Reference
KNN	80	0.36	290	0.04	130	—	—	—	6
KNN (HP)	160	0.45	420	—	240	—	—	—	67
KNN (SPS)	148	0.39	559	0.036	—	395	220	—	69
KNN	160	—	586	—	—	415	209	—	18
KNNL	235	0.44	534	0.04	—	465	75	1.021	32
KNNL	280	0.48	760	—	—	475	60	1.021	39
KNNL	324	—	—	—	—	475	60	1.025	40
KNNS	204	0.47	850	0.02	150	300	130	1.006	49
KNNT	270	0.44	917	0.005	455	208	87	—	80
KNNT	151	0.42	700	0.03	75	328	170	—	48
KNNLT	190	0.46	920	0.024	—	310	52	—	33
KNNLT	206	0.38	540	0.02	140	—	—	1.014	81
KNNLT	250	0.37	440	0.043	—	—	—	—	82
KNNLT	230	0.43	600	—	—	—	—	—	83
KNNLT	208	0.48	1146	0.02	—	320	51	1.014	84
KNNLT	268	0.46	666	0.011	—	435	45	1.014	85,86
KNNLT	276	0.46	650	0.03	18	445	69	1.021	87
KNNLT	263	0.53	1290	0.019	—	325	40	1.016	88
KNNLT (SPS)	243	0.46	1240	0.023	85	—	—	1.016	89
KNNLS	265	0.5	1380	0.02	40	368	35	—	44,90
KNNLS	286	0.51	1372	0.019	—	385	30	—	91
KNNLS	212	0.46	1335	0.028	—	358	48	1.014	51
KNNLS	280	0.49	1464	0.023	—	364	25	1.014	23
KNNLS	240	0.42	1000	0.03	—	—	—	1.014	92
KNNLTS	416	0.61	1570	—	—	253	—	1.013	5
KNNLTS	345	—	1175	—	—	260	25	—	93
KNNLTS	252	0.42	1125	0.03	—	275	—	—	94,95
KNNLTS	335	0.53	2064	—	41	291	70	1.008	5,96
KNNLTS	250	0.45	1352	0.03	—	348	74	1.012	97
KNNLTS	263	0.45	1473	0.02	—	353	50	1.015	98
KNNLTS	306	0.48	1327	0.025	—	337	30	1.014	22
KNNLTS	252	0.42	1503	0.025	—	—	—	1.010	99
KNNLTS	264	0.385	689	0.02	—	338	—	1.014	100
KNNLTS	400	0.54	2500	—	—	230	—	1.004	101

Table II. Properties of KNN-Based Piezoceramics

Material	d_{33} (pC/N)	k_p (%)	ϵ_r	$\tan \delta$	Q_m	T_C (°C)	T_{O-T} (°C)	c/a	Reference
KNN-AgNbO ₃	186	0.42	495	—	—	355	165	—	102
KNNLT-AgSbO ₃	265	—	1000	—	—	235	40	1.009	103
KNN-BiScO ₃	210	0.45	1100	—	—	340	75	—	104
KNN-BiScO ₃	253	0.48	1150	—	—	351	100	1.009	57
KNNL-CZ	216	0.429	1088	—	—	330	—	1.016	53
KNNLT-CZ	320	0.47	1900	0.02	—	195	50	1.007	105
KNNLS-CT	210	0.49	1170	0.02	—	330	-15	—	106
KNNLS-CT	200	—	—	—	—	—	—	—	107
KNNLTS-CT	263	0.50	1490	0.03	—	315	—	—	108
KNN-BT	225	0.36	1058	—	—	—	—	—	109
KNN-BT	193	0.43	1100	0.02	—	314	—	1.010	54
KNNLTS-BZ	365	0.45	3400	—	—	175	25	1.006	110
KNN-BZT	234	0.49	1191	0.01	—	318	50	1.014	55,111
KNN-ST	220	0.4	1147	—	72	—	—	1.010	112
KNN-BiFeO ₃	173	0.4	905	0.04	—	390	—	1.008	113
KNNLT-BiFeO ₃	200	0.36	1140	0.018	80	305	—	1.013	114
KNN-BNT	195	0.43	900	—	—	375	—	1.016	115,116
KNN-BKT	251	0.49	1260	0.03	—	376	—	1.014	117

could contribute to d_{33} as extrinsic contributions. However, it is argued whether the d_{33} should increase monotonously with reduction of c/a because once spontaneous polarization is too small it cannot induce effective piezoelectric response. In some materials like BiFeO₃-PbTiO₃, there is an optimal value for c/a ¹¹⁹; however, it is not observed in KNN due to certain undiscovered reasons.

(2) Domain Engineering

Ferroelectric domains are essential to piezoelectric materials, which could link with both intrinsic and extrinsic parts. The former refers to the deformation of a single domain, whereas the latter can be related with the examples of domain rotations, movement of domain walls, etc. It should be clarified that the terminology “domain engineering” here is the

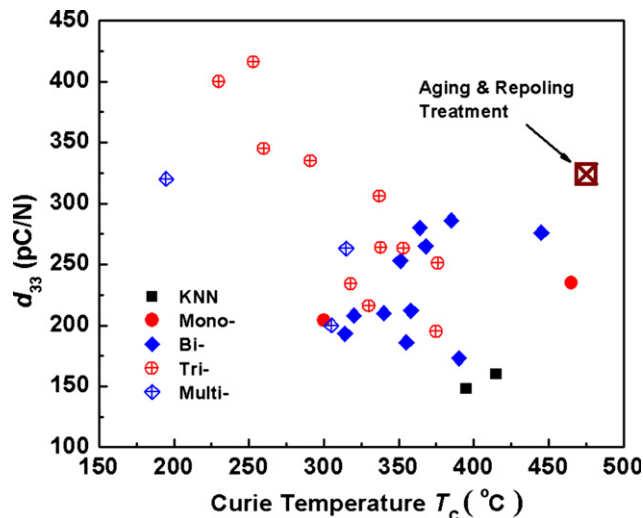


Fig. 6. Curie temperature-dependent piezoelectric coefficient of KNN piezoceramics modified with desirable dopants. Mono-, Bi-, Tri-, and Multi- refer to adding one, two, three and more kinds of dopants in pure KNN matrix, respectively.^{5,22,23,32,40,44,49,51,53–55,57,67,69,84,87,91,93,96,98,100–102,104,105,108,113–115,117}

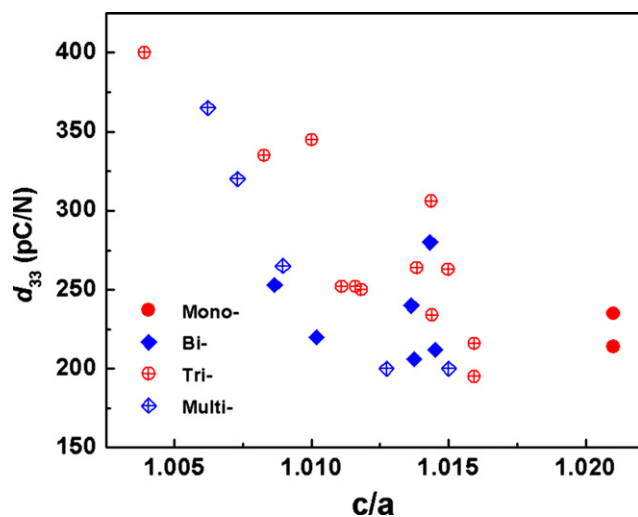


Fig. 7. Piezoelectric coefficient of KNN piezoceramics doped with various species as a function of c/a ratio.^{22,23,32,49,51,53,55,57,81,92,93,95–101,103,105,107,110,112,114,115}

alignment of domains in polycrystals rather than special engineered domain structures in single crystals.^{120,121} The recently discussed contribution from “charged domain walls” is also not concerned in the present review.^{60,122,123}

Domains can be generally aligned by applying a high electric field up to several thousand voltages per millimeter, namely poling process, which is considered as an important process to endow ferroelectric ceramics with macroscopic piezoelectric responses. Inferior ferroelectric response is obtained by insufficient poling, which is caused by various factors such as defects. The influences of poling conditions on the dielectric and piezoelectric properties of KNN-based ceramics has been a subject of recent studies,^{35,124–129} most of which point out that poling temperature is of significant importance, with respect to individual compositions as well as phase structures. Nevertheless, poling process could also be utilized to manipulate interior dipole movement, which might be tuned to favor piezoelectric performance.^{40,42,130}

For an optimized Li-doped composition, its piezoelectric coefficient d_{33} can be increased up to 324 pC/N even from an

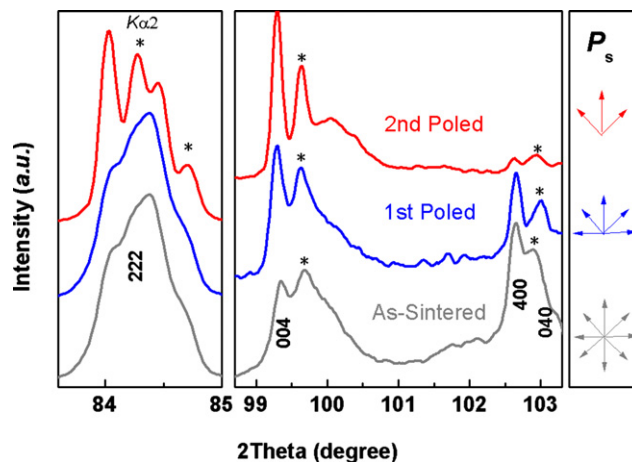


Fig. 8. X-ray diffraction patterns of (222) and (004) lattice plane series for the sample $0.92(\text{Na}_{0.535}\text{K}_{0.48})\text{NbO}_3\text{--}0.08\text{LiNbO}_3$, both before and after the first poling, as well as after the second poling. The evolution of P_s inside domains is schematically illustrated in a separate box on the right-hand side, assuming that the electric field applied during poling is parallel to $\langle 001 \rangle$ direction.⁴⁰

already large value of 190 pC/N by means of repoling after room-temperature aging⁴⁰; such a high d_{33} value is only reachable for KNN ceramics with complicated modifications (see Fig. 6). It should be noted that this treatment is most effective in the suitable phase structure with orthorhombic–tetragonal two-phase coexistence, whereas deviations toward either end members lead to ineffectiveness, or even deterioration. The large increment of d_{33} is considered due to improvement of P_r after the special treatment, which is evidenced by the apparent domain switching deduced from XRD patterns, as indicated in Fig. 8. It should be noted that this phenomenon is quite unique for KNN that no similar effects have been reported in other systems. One possible mechanism corresponding to the details of spontaneous polarization change in the domain level has been proposed, concerning the combined effect of migration of oxygen vacancies, and interaction between defect dipoles and spontaneous polarization inside domains.⁴⁰

(3) Enhancement of Temperature Stability

As mentioned previously, the room-temperature piezoelectric properties in KNN-based ceramics can be largely improved by proper chemical modifications, which are actually due to the tetragonal–orthorhombic PPT point (T_{O-T}) shifting downward from around 200°C toward room temperature, namely PPT effect. Although sharing similarities with MPB in terms of property enhancement due to two-phase coexistence, the PPT effect cannot help but suffer from problems of temperature instability.^{36,44,88,93,105,106,108,128,131–137} For example, Zhang *et al.* reported that for LiSbO_3 -doped KNN ceramics, the normalized strain d_{33}^* dropped from 355 pm/V at room temperature to around 250 pm/V at 50°C and the planar coupling factor k_p also decreased linearly with increasing temperature above room temperature.⁴⁴ Hollenstein *et al.* demonstrated that the piezoelectric coefficient d_{31} and coupling factor k_p decreased up to 30% of their initial value after the first heating cycle of up to 140°C for Li-modified KNN.¹³¹ Akdoğan *et al.* found that for LiTaO_3 and LiSbO_3 codoped KNN ceramics, the spontaneous polarization P_s , permittivity, and piezoelectric coefficient d_{33} all peak in a narrow temperature range at 25°C–31°C.⁹³

To improve the temperature stability, two possible solutions have been adopted so far: shifting the T_{O-T} well below room temperature by chemical modifications,^{106,108,134} or producing highly textured ceramic samples.^{5,136} However, the

former actually enhances the temperature stability by inducing a single tetragonal phase at room temperature by bringing T_{O-T} down below room temperature, which eventually sacrifices piezoelectric activity (compared with those compositions benefited by PPT effect at room temperature) to some extent; whereas the latter requires complex synthesis process relatively incompatible with mass production.

Ideally, if a real MPB-like phase boundary could be established in the KNN system, the temperature sensitivity issues might be solved completely. It has been considered as a solution to induce a tetragonal-rhombohedral phase boundary. From the phase diagram of pure KNN, the rhombohedral phases are stable below 100 K when K/Na ratio is close to 50/50.^{20,25} Recently, it was reported that the rhombohedral phase could be shifted toward room temperature while the orthorhombic phase gradually disappears via chemical modification, resulting in seemingly rhombohedral-tetragonal phase coexistence at room temperature.^{53,110,138–140} However, it turns out to be different from the MPB in PZT and still suffers from temperature instability issues. Previously, we added BiFeO₃ (rhombohedral) into modified KNN (tetragonal) matrix; nevertheless, it was found that small amount of BiFeO₃ induced a pseudocubic structure in the matrix, which is not clearly identified but shows only slight deviations from the cubic phase.^{114,141}

Recently, we have demonstrated that it is possible to smooth the deviations between tetragonal and orthorhombic phases, resulting in a stable electric field-induced strain behavior over a wide temperature range, as shown in Fig. 9.¹⁰⁵ In this example, CaZrO₃ was introduced for the structure modification, together with commonly used dopants such as Li and Ta. It is not the authors who firstly introduced CaZrO₃ into KNN system; however, temperature-dependent strain behavior was not discovered before.^{53,142} On the other hand, it is expected that CaZrO₃ is not the only species that can improve temperature stabilities in KNN, while the smoothing between tetragonal and orthorhombic phases is important.

However, challenges still exist. The introduction of CaZrO₃ certainly improves temperature insensitivity of electric field-induced strain behavior but not the small signal piezoelectric coefficient d_{33} .¹⁰⁵ Even for textured KNN ceramics, the enhanced temperature stability demonstrated by Saito *et al.* is focused on actually unipolar strain behavior, not the small signal d_{33} .⁵ It is considered that the existence of PPT point around room temperature may deteriorate spontaneous polarization during heating/cooling, which is very important for possible solution to this problem.

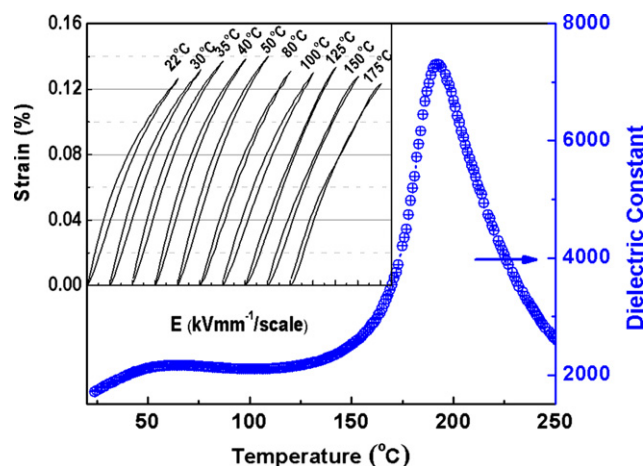


Fig. 9. Dielectric permittivity and loss of CaZrO₃-modified KNN ceramics as a function of temperature; the inset figure is temperature dependence of unipolar strain in CaZrO₃-modified KNN ceramics.¹⁰⁵

IV. Sintering of KNN-Based Ceramics

(1) Normal Sintering

It is known that KNN ceramics have been accepted as one of the promising lead-free materials since 1950s.^{6,143} However, at first it was a big challenge to obtain highly dense bulk samples by normal sintering. Later in 1962, an advanced fabrication technique of HP was introduced by Jaeger and Egerton, resulting in high bulk density (~95% of theoretical density) with enhanced properties, e.g., high Curie temperature ($T_C \sim 420^\circ\text{C}$), large d_{33} (~160 pC/N) and k_p (~45%).^{67,68} Nevertheless, normal sintering has still been preferred in the community due to its significant advantages in shaping flexibility and cost efficiency. It should be noted that the sintering temperature of KNN-based ceramics is relatively high (over 1000°C), considering its melting point at 1200°C, although depending on compositions; thus, a large degree of loss in alkali elements and consequent compositional fluctuation is generally unavoidable. Several research groups have studied the volatilization behaviors of KNN-based ceramics.^{33,50,81,144–148} Our previous work confirmed that serious volatilization happened over 1000°C (Fig. 10), which was determined by inductively coupled plasma (ICP) analysis. It is suggested that Li, Na, and K evaporate at different levels, in agreement with the reports by Park *et al.* and Zhang *et al.*^{37,148–150} Later on, Cheng *et al.* investigated the distribution of alkali metal contents along the depth direction from the surface by semiquantitative X-ray photoelectron spectroscopy measurement, revealing that the volatilization of Li is the most serious, particularly at the ~20-μm-thick outmost surface, whereas the volatilization of Na and K elements becomes less severe from the as-sintered surface to the interior part.¹⁵¹

The volatilization problem could seriously deteriorate piezoelectric properties. Supposing there is no volatilization in KNN-based ceramics, the highest d_{33} is expected when the density peaks, which is not the case in reality, as shown in Fig. 11. It should be noted that the degree of volatilization of Li, Na, and K is all temperature dependent, but different in extent. The A-site compositions of KNN are critically important in tailoring the phases and hence properties, which is quite different from the case of PZT. It is well-known that the properties of PZT are strongly dependent on the Zr/Ti ratio on the B-site.^{144,81,146} Although various factors such as density influence piezoelectric performances, the d_{33} of KNN-based ceramics is mainly dependent on the PPT temperature, which has been verified by many studies.^{32,43,50,60,104,153,154} The determination of the optimal sintering temperature for KNN with highest d_{33} is only meaningful when the T_{O-T} is maintained at room

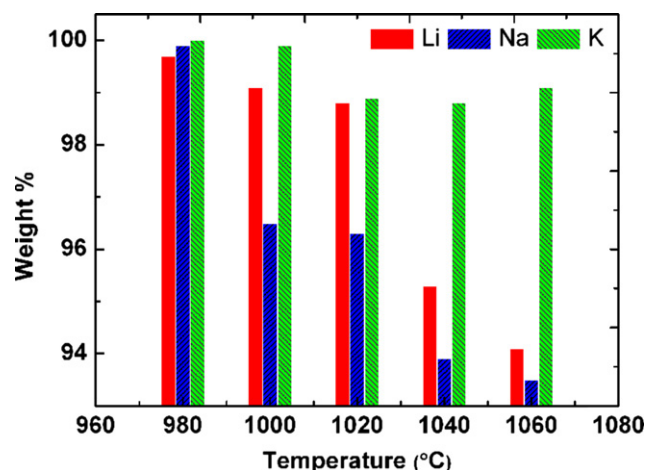


Fig. 10. The degree of volatilization of alkali elements (Li, Na, K) as a function of sintering temperatures in KNN-based ceramics.¹⁵²

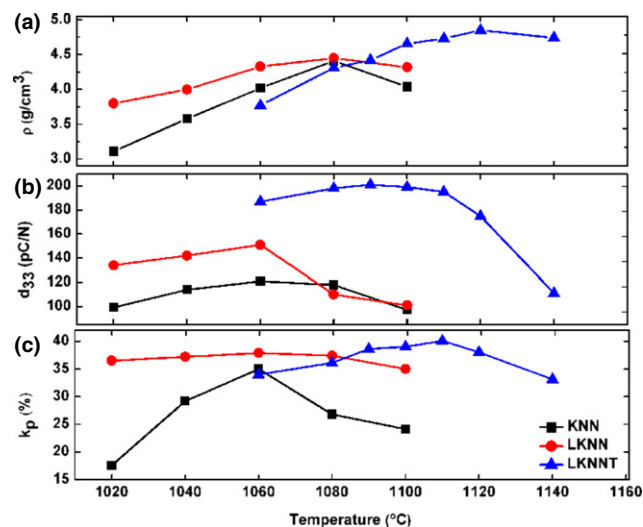


Fig. 11. Plots summarizing properties of K_{0.5}Na_{0.5}NbO₃ (KNN), LiNbO₃-modified KNN (LKNN), Li- and Ta-modified KNN (LKNNT) ceramics as a function of temperature: (a) Bulk density versus sintering temperatures; (b) Piezoelectric coefficient d_{33} versus sintering temperatures; (c) Planar mode electromechanical coupling coefficient k_p versus sintering temperature.¹⁵⁸

temperature. As noted, increasing sintering temperature usually results in the enhanced densification. However, sintering above certain temperature ends up with the loss of A-site elements, eventually leading to a shift in T_{O-T} and consequent degradation in d_{33} in spite of an apparent increase in the density. As well, another commonly encountered problem due to high volatilization is the appearance of secondary phases, such as tungsten bronze-structured K₃Li₂Nb₅O₁₅, Li₃NbO₄, and K₄Nb₆O₁₇.^{38,50,64,109,145,155}

Given the aforementioned negative consequences of the volatilization problems, several innovative processing techniques have been proposed. For example, Zhen *et al.*⁸¹ proposed a “Double Crucible” method, where the target sample is separated from the atmospheric powder by using an additional crucible, made of KNN ceramics during calcination. On the other hand, many researchers suggested that adding more alkali carbonates into starting powders can keep the deviation in the Li/Na/K ratio within a reasonable range, and hence improve the piezoelectric/ferroelectric properties.^{40,41,145,156,157}

In addition to volatilization, another feature usually observed during sintering of KNN-based ceramics is the abnormal grain growth (AGG). Piezoelectric properties decrease in the highly dense ceramics with AGG and *new core-shell* structure.¹⁵⁹ Cubic-like faceted grains in contrast to polyhedral grains observed in PZT represent the typical microstructures of well-sintered KNN ceramics, which is considered to be a consequence of liquid-phase sintering.¹⁶⁰ Thus, the observed facets in KNN grains are [001] series crystallographic planes with the lowest surface energy.^{161,162} Generally, the sintering temperature of KNN is relatively high near its melting point, resulting in ineluctable appearance of liquid phase. As a consequence, possibility of solid-state sintering is significantly reduced.

Appearance of liquid phase during sintering promotes the AGG behavior as well as secondary phase.^{37,163–166} Generally, grain growth is classified into two types: normal and abnormal (exaggerated) grain growth.¹⁶⁰ Normal grain growth is characterized by unimodal distribution of relative grain sizes with time; in contrast, AGG involves some unexpectedly large grains out of a matrix of fine grains, showing a bimodal grain size distribution.¹⁵⁹ Here, the grain growth behavior of KNN will be addressed in terms of the microstructure changes as a function of sintering temperatures.

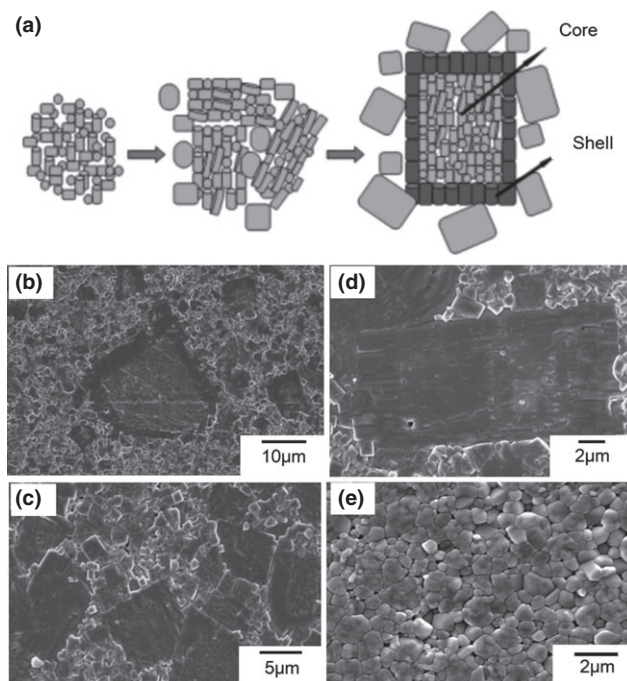


Fig. 12. New core-shell structure in coarse grains: (a) Schematic diagram showing the formation mechanism of a new core-shell structure: from left to right, Step 1: randomly distributed nanosized grains; Step 2: self-assembled aggregation clusters of nanosized grains; and Step 3: typical new core-shell structure of a coarse grain; (b) (K, Na)NbO₃ sintered at 1100°C; (c) (Li_{0.85}K_{0.44}Na_{0.52})(Nb_{0.85}Ta_{0.15})O₃ sintering at 1120°C; (d) final formation of a core-shell structure when sintered at 1100°C in (K, Na)NbO₃ samples at a high magnification; (e) SEM image of thermally etched surface of (Li, K, Na)NbO₃ ceramics doped with 2 mol% CaZrO₃ sintered at 1020°C.

For the KNN-based ceramics sintered at elevated temperatures, the microstructure evolution is similar to a grain coalescence process [Fig. 12(a)]. First, a homogeneous microstructure with quadrate grain shape powders (~500 nm to ~1 μm) shows up at relatively low sintering temperature. Then when the sintering temperature passes through a critical point, the grain coalescence suddenly appears in certain part of grains, which exaggerated to 2–5 μm in size, as shown in Figs. 12(b)–(d); meanwhile, other grains still retain their original shapes and sizes. When the sintering temperature further increases, the liquid phase may be formed which accelerates the particle redistribution because of the enhanced diffusion process.

Again, similar to the volatilization problem, AGG is normally not avoidable. Therefore, achieving as high density as possible without the formation of liquid phase is always preferred. With the aid of proper chemical modifications and elaborately controlled sintering procedures, solid-state sintering of KNN could be achieved, as shown in Fig. 12(e). As well, it should be noted that controlling the particle size of starting powders, which refers to those obtained after calcination, is quite important. Many milling strategies have been investigated, such as high-energy ball milling, prolonging ball-milling time.^{86,155,167} It was revealed that prolonging the milling period of calcined powders would significantly reduce the particle size, promoting the densification, resulting in fine grain size (~500 nm) with less porosity, as revealed in Fig. 13. This is easily understood by the fact that fine particle sizes help lowering the sintering temperature. In addition, controlling heating and cooling rate during sintering of KNN-based ceramics requires a special attention to guarantee the final properties which has been demonstrated by Zhang *et al.*¹⁴⁸ A slow cooling rate allows the compositional

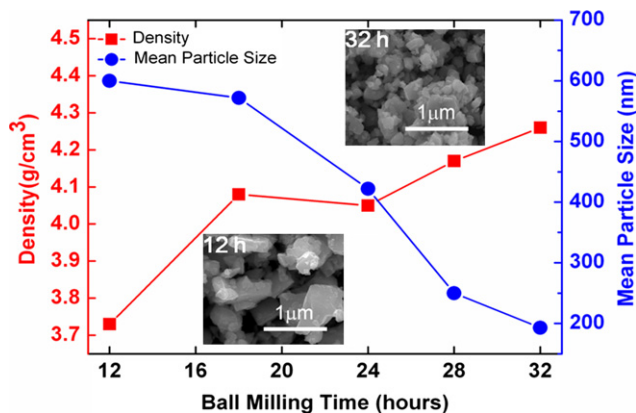


Fig. 13. The density and mean particle size of $\text{Li}_{0.058}(\text{Na}_{0.52}\text{K}_{0.48})_{0.942}\text{NbO}_3$ from conventional solid-state sintered powder as a function of ball-milling time. Inset SEM images represent the calcined powder after ball milling of 12 and 32 h.¹⁵⁶

equilibrium to establish, having K and Na distributed homogeneously to enhance piezoelectric properties.

An interesting phenomenon reported by Choi *et al.*¹⁴² and Zhang *et al.*¹⁴⁸ is the chemical segregations appearing in some cases, leading to nonuniform microstructures with degraded electrical properties. In these works, a core-shell structure was observed and investigated by transmission

electron microscopy (TEM) and chemical segregations were clearly detected by EDS (energy-dispersive X-ray spectroscopy) as presented in Fig. 14. For the $\text{K}_{0.5}\text{Na}_{0.5}\text{NbO}_3$ -Li-TaO₃-BiScO₃ (KNN-LT-BS) pseudoternary system, core-shell structure exists in the fine grain size samples ($\sim 0.5 \mu\text{m}$). It was found that Na and K uniformly were distributed in a grain, but Nb was present preferentially in the core while Bi, Ta, and Sc were rich in the shell.^{155,167} This concentration difference in between core and shell regions was shown to result in twin dielectric peaks at high temperature.^{32,155} It is interesting to notice that this core-shell structure was highly stable against prolonging high-temperature annealing.¹⁶⁸

Due to the difficulty in avoiding AGG and in sintering KNN at solid state, various sintering aids have been considered. An ideal sintering aid for the sintering of KNN should be the one that minimizes the effect of AGG, while decreases sintering temperatures as much as possible. For the enhancement of the properties in KNN-based materials, high compact density and relatively low sintering temperature are always desirable. Adding sintering aids that effectively reduce the sintering temperature by forming a small amount of liquid phase would be useful in achieving high density with a minimum compositional fluctuation. Excess alkali metal contents (such as Li_2O and Na_2O) have been considered as useful sintering aids in many reports, although some of them actually act also as a compensation for the volatilization. The frequently used sintering aids in KNN-based lead-free piezoelectric ceramics are summarized in Table III. It is shown that with the help of sintering aids, the relative

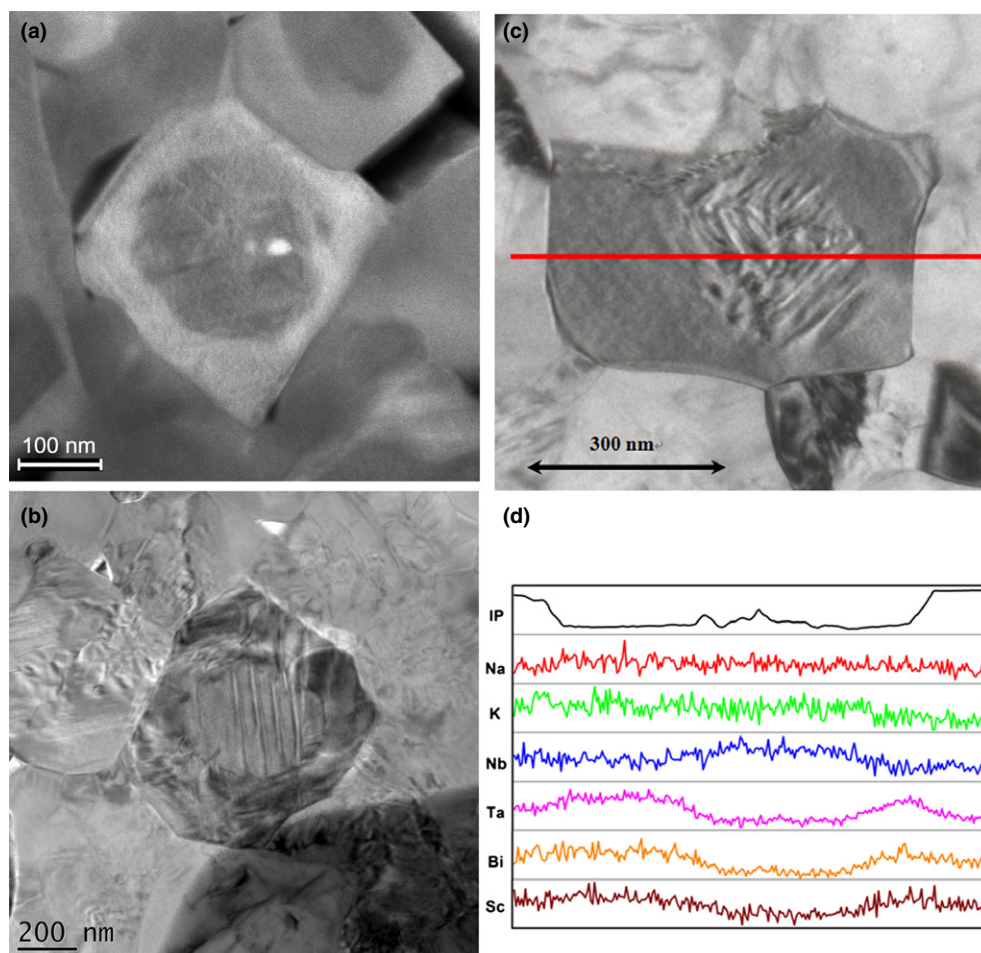


Fig. 14. TEM studies of KNN-LT-BS lead-free ceramics. For the composition of $0.93(\text{K}_{0.5}\text{Na}_{0.5})\text{NbO}_3$ - 0.05LiTaO_3 - 0.02BiScO_3 ceramics: (a) HAADF (high-angle annular dark-field) image, where the contrast indicates chemical concentrations; (b) bright-field image, in which the core and shell structure was vividly revealed. For the $0.88(\text{K}_{0.5}\text{Na}_{0.5})\text{NbO}_3$ - 0.07LiTaO_3 - 0.05BiScO_3 compositions: (c) bright-field TEM image; (d) energy-dispersive X-ray spectroscopy line scanning across the grain corresponding to the red line in (c), IP (intensity profile) was used to position the elemental line scans. The hump for Nb indicates elemental concentration, *vice versa*.^{155,167}

Table III. KNN-Based Materials with the Sintering Aids

Base	Dopant	Sintering temp.(°C)	Density (g/cm ³)/theoretical(%)	d_{33} (pC/N)	k_p	T_C (°C)	T_{O-T} (°C)	Reference
KNN	—	1050–1100	4.26/94	97	0.29	420	200	32
	Mg ²⁺	1115	4.11/92.2	—	—	406	179	169
	Ca ²⁺	1115	4.23/93.8	95	—	398	171	169
	Sr ²⁺	1115	4.33/96	95	—	397	173	169
	Ba ²⁺	1115	4.24/94.21	—	—	404	191	169
	LiF	900–980	4.90/—	—	0.30	235	30	170
	ZnO	1050	4.28/—	123	0.42	420	200	171
	CuO	1060–1120	—/95	82	0.39	421	226	172
	GeO ₂	1000	4.31/95.6	120	0.38	—	—	173
	BaTiO ₃	1030–1060	—/90	225	0.36	—	—	109
	(Bi,K)TiO ₃	1120–1140	4.38/97	251	0.49	376	200	116,117
	LiSbO ₃	1080	4.28/95	286	0.51	385	25	44,51,90,91
	Ba(Zr,Ti)O ₃	1100–1160	—/97	243	0.49	318	—	55,174
	K ₄ CuNb ₈ O ₂₃	1065–1150	4.51/98	103	0.40	—	—	175
	K _{1.94} Zn _{1.06} Ta _{5.19} O ₁₅	1070	4.53/99.2	126	0.42	400	190	176
	MnO ₂ –K _{5.4} Cu _{1.3} Ta ₁₀ O ₂₉	1110	4.65/—	90	0.40	404	208	111
	LiSbO ₃ –CaTiO ₃	1120–1180	4.55/99	210	0.61	330	–15	106
Other (K, Na)NbO ₃ -based ceramics								
KNN–BZT	MnO ₂	1100–1170	4.45/97	234	0.49	318	—	174
KNNS	CuO ₂	960	—/94.8	122	0.41	320	120	177
KNNS	MnO ₂	1100–1140	4.53/—	213	0.48	320	150	49
KNNLT	Li ₂ O	1000	4.41/96%	250	0.37	—	—	23
KNLNT	Li ₂ O	1000	—/97%	290	0.46	—	—	24
KNNLN	Na ₂ O	950	4.38/97%	280	0.48	50	50	25
KNNLT	Na ₂ O	1040	—	230	0.43	—	—	26

Conventional solid-state synthesis method.

density can be increased over 94% of theoretical density while d_{33} and k_p are improved at the same time.

(2) Spark Plasma Sintering (SPS)

To achieve much enhanced piezo/ferroelectric properties, special sintering techniques are required, such as HP and SPS. In principle, SPS is similar to HP, although it has its unique features, e.g., low sintering temperatures, stress-assisted densification process, large current shock effect, etc. Although it is still argued whether the plasma is truly involved during sintering or not, SPS has been successfully employed to manufacture high-performance KNN-based ceramics.

It is generally considered that SPS is an ideal tool for sintering KNN-like ceramics, which is highly volatile while difficult to be sintered at solid state. The SPS apparatus generally consists of a uniaxial pressing system, in which water-cooled punches serve as electrodes. The powders are placed inside a graphite die and then the chamber is evacuated before sintering. After power on, the pulsed DC generator, which supplies a huge current up to 1000 A and enormously enhanced heating speed, will work together with complicated pressure-, position-, and temperature- controlling systems. This technique utilizes a synergic effect between a considerably high pressure (20–100 MPa) and fast heating rate (100°C/min–300°C/min), realizing highly dense bulk ceramics with fine grains within several minutes.^{69,70,89,178,179} For example, pure KNN ceramics have been successfully sintered at a temperature as low as 920°C, which is about 200°C lower than that for normal sintering. Up to ~99% of density was achieved with an enhanced d_{33} up to 148 pC/N. It is noticed that normal sintering usually results in 92% of density with d_{33} of around 80 pC/N. Interestingly, the permittivity was also observed to be enhanced by 50%. It is considered that the enhanced piezo- and dielectric properties are due to the high density, reduced grain sizes (200–500 nm), and better stoichiometry. The usefulness of this technique was further verified by the successful synthesis of KNN ceramics with dopants, such as Li and Ta, the

enhanced properties, as summarized in Fig. 15 and Table IV.^{87,89}

Another advantage using SPS is the significantly improved mechanical properties, as depicted in Fig. 16. Although both samples sintered using normal method and by SPS exhibit the brittle fracture behavior in MSP (modified small punch method) load-displacement test,⁸⁹ the indentation test suggests that the SPS-sintered samples are much harder than their counterparts from normal sintering. The MSP load is proportional to the displacement before the fracture point followed by an abrupt load decline in the curve, which can be attributed to the stress release via crack propagation. Subsequently, accompanied by relatively larger displacements, the MSP load of the sample increases to the maximum value, and then ultimate fracture occurs. The fracture loads, which implies crack initiation in the sample loaded, are $P = 28.9$ and 23.1 N for the samples from SPS and from normal sintering, respectively. It is suggested that the enhanced mechanical properties of SPS-sintered ceramics also result from the improved densification and fine-grained microstructures.

It is evident that enormous amount of researches have been performed to enhance the functional properties using the SPS technique, which turns out successful in most cases as demonstrated in the present KNN-based lead-free piezoceramics. Nevertheless, it should be pointed out that the SPS technique may be more suitable for laboratory scale rather than for industry scale because SPS lacks the degrees of freedom in processing apart from its high processing cost as compared with normal sintering.

V. Promising Applications for Transducers and Energy Harvesting

Advances in fabrication and characterization of KNN-based materials have resulted in the development of their applications for electronic devices and energy.^{12,111,180–183} Several research groups have presented related studies, many of which are focused on sensors, transducers, and actuators.^{181,184–195} For example, Xiao *et al.* utilized the

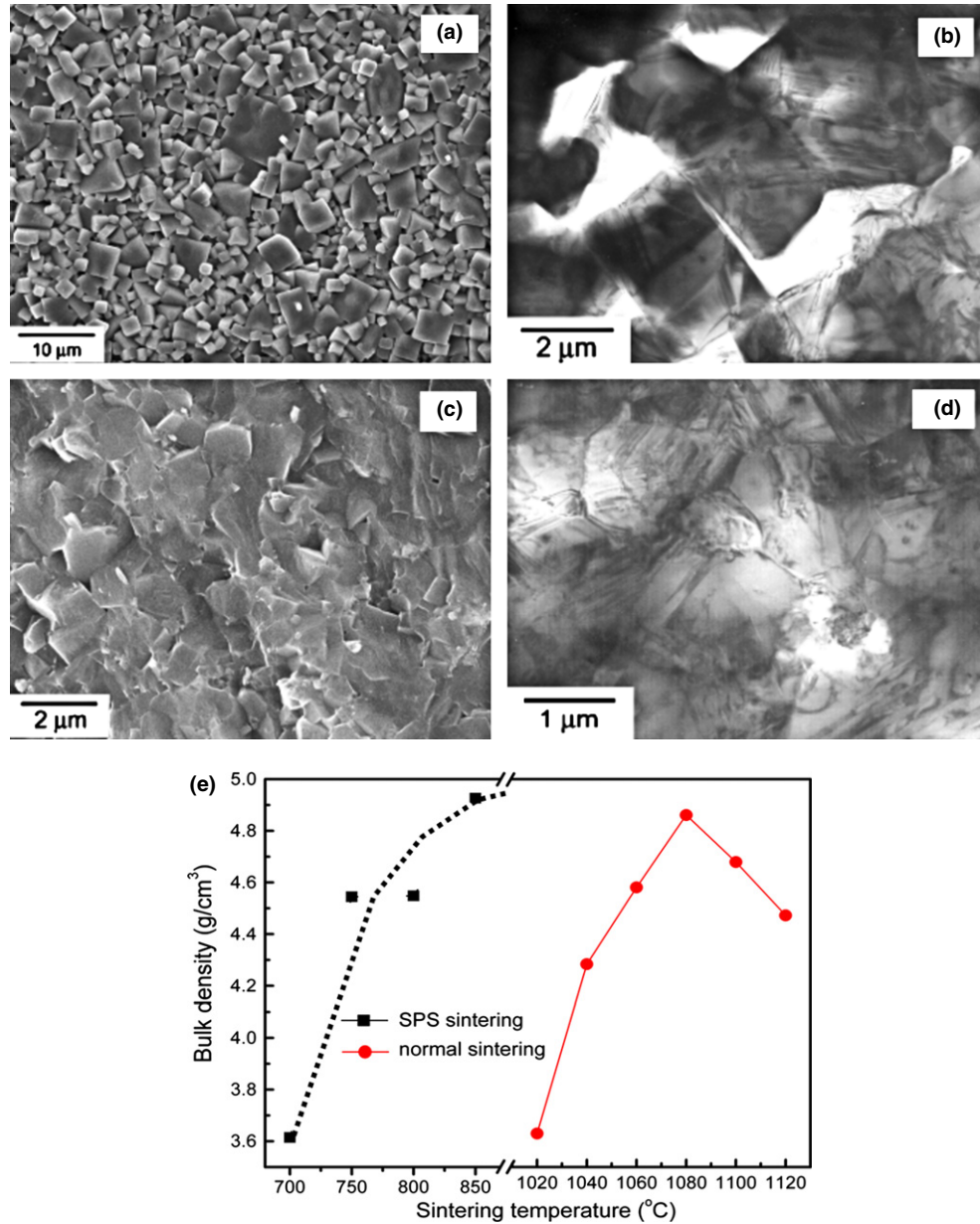


Fig. 15. The composition of $(\text{Na}_{0.535}\text{K}_{0.485})_{0.95}\text{Li}_{0.05}(\text{Nb}_{0.8}\text{Ta}_{0.2})\text{O}_3$ ceramics: (a) SEM image, (b) TEM image from normally sintered ceramics; (c) SEM image, (d) TEM image of spark plasma sintering (SPS)-sintered ceramics; (e) bulk density as a function of sintering temperature in comparison with SPS method and conventional solid-state synthesis.⁸⁹

Table IV. Properties of $(\text{Na}_{0.535}\text{K}_{0.485})_{0.95}\text{Li}_{0.05}(\text{Nb}_{0.8}\text{Ta}_{0.2})\text{O}_3$ Ceramics with Normal Sintering and Spark Plasma Sintering (SPS)

[†]	d_{33} (pC/N)	k_p (%)	Q_m	P_r (μC/cm ²)	E_C (kV/mm)	ϵ_r (1 Hz)	$\tan \delta$ (1 Hz)	σ_f (MPa)
Normal	176	30.8	82	—	—	1113	0.032	67.9
SPS	243	46.1	85.1	26.5	1.74	1240	0.023	102.84

[†]Conditions: Normal sintering, power calcined at 750°C for 5 h, isostatic press at 200 MPa, sintering at 1080°C for 2 h with heating/cooling rate at 10°C/min in the air⁸⁷; SPS, evacuated chamber ($\sim 10^{-2}$ Pa), sintering at 900°C for 10 s with soaking rate at 100°C/min at constant 50 MPa, annealed in the air at 900°C for 6 h.⁸⁹

composition $(\text{K}_{0.5-x}\text{Li}_x)\text{Na}_{0.5}(\text{Nb}_{1-y}\text{Sb}_y)\text{O}_3$ (abbreviated as KLNNS_{x-y} , $x = 0-4$ mol% and $y = 0-8$ mol%) for a successful buzzer prototype, satisfying the requirement for practical applications.¹⁸³ A pin transducer, which consisted of two types of KNN-based ceramics, was obtained by Lam *et al.* with a purpose of a structural monitoring application.¹⁸⁰ Another KNN-based ceramics was prepared and used as a voltage transformer by Guo *et al.*¹⁸⁴ Regarding the activities from industry, Murata Manufacturing Co. Ltd., (Nagaokakyo, Kyoto, Japan) demonstrated that KNN-based multilayer piezoelectric ceramics could achieve comparable

displacement as PZT-based multilayer actuators, while what is more important is that nickel could be employed as inner electrodes (in a reducing atmosphere with oxygen partial pressure of 10^{-11} – 10^{-10} MPa).⁸ The TDK Corp. (Narita, Chiba, Japan) prepared $(\text{K}, \text{Na}, \text{Li}, \text{Ba}, \text{Sr})(\text{Nb}, \text{Ta}, \text{Zr})\text{O}_3$ (abbreviated as ANSZ) ceramics with good thermal reliability. The change in d_{33} was comparable with that of PZT ceramics after 1000 cycles of thermal shock within the temperature ranging from -55°C to 125°C , which made a promising candidate for actuator applications.¹³⁵ A miniature piezoelectric ultrasonic motor (USM; Tokyo Institute of Technology, Tokyo, Japan)

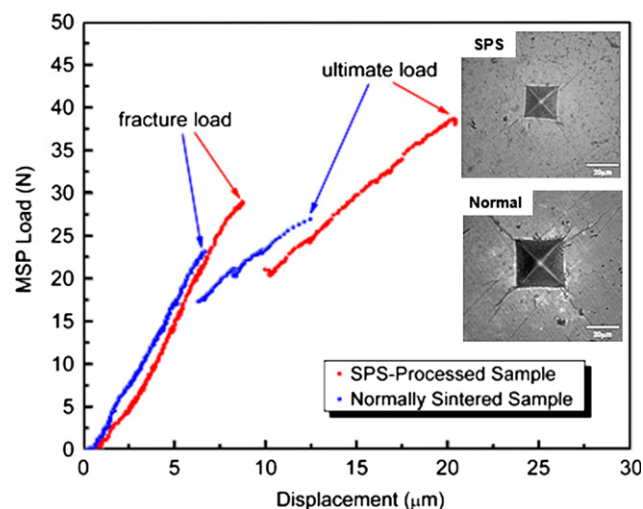


Fig. 16. Modified small punch (MSP) load–displacement curves for spark plasma sintering processed and normally sintered $(\text{Na}_{0.535}\text{K}_{0.485})_{0.95}\text{Li}_{0.05}(\text{Nb}_{0.8}\text{Ta}_{0.2})\text{O}_3$ ceramics. Inset SEM images of indentation test were taken after MSP load–displacement measurement.⁸⁹

was also fabricated using KNN ceramics, which can be driven in the shearing and bending vibration modes, also demonstrating good potential for actuator applications.¹⁹⁶

The aforementioned examples indicate that it is possible to substitute the PZT-based piezoceramics, maybe not fully but at least partially in the near future. The most controversial weakness of KNN lies in two parts, which are the inferior piezoelectric properties (compared with PZT) and the sensitive processing issues. As mentioned in the Introduction, KNN is not necessarily required to substitute all the PZT family as an “all-round player,” while it does provide competitive performances against at least certain types of PZT, e.g., “hard” PZT (a typical representative is PZT-8; BOCHUANG CERAMIC, INC., Shanghai, China). Due to the vast applications of these specific PZT types, environmental issues could be improved step by step, where KNN-based piezoceramics play an important role. On the other hand, it is considered that the processing arguments are not essential due to the significant difference between lab-level preparation and industry implementation, whereas the latter is more advanced in terms of processing facilities as well as production control, which is already implied by the several successful stories from companies.

Despite the moderate piezoelectric properties compared with PZT, the KNN-based ceramics do possess certain distinct features, which can manifest their superiorities. This aspects will be discussed to a degree with the following two examples,^{190,197–199} although these do not limit the attractive application regimes for KNN-based piezoceramics. Once KNN-based piezoceramics are further understood and recognized by the academia as well as industry, it is believed that more and more advantages of using KNN-based piezoceramics will be revealed.

(1) KNN-Based 1–3 Composite

Generally, KNN possesses lower permittivity ϵ_r as well as mechanical quality factor Q_m , when compared with PZT ceramics. Nevertheless, these features could be favorable for transducer-oriented applications, such as ultrasonic imaging.²⁰⁰ Assuming a given composition of piezoceramics, the coupling factor for these kinds of applications can be further enhanced if the material forms 1–3-type composite together with polymers. The 1–3 piezoceramic/polymer composites are applied especially for high-frequency hydrophones and biomedical imaging transducers.^{182,201,202} Normally, two component parts are

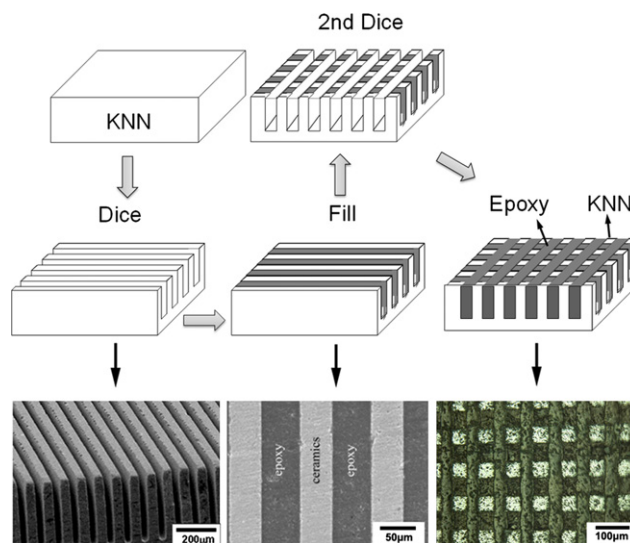


Fig. 17. Procedures of dicing and filling process, and the SEM and optical images of the related steps. The “dice-fill” process: a KNN disk was diced to make a set of grooves and filled with epoxy; the 2nd set of grooves was diced after the epoxy was cured at 30°C–40°C for over 24 h; after that, the epoxy was filled and cured again; grinding the sides of the samples until the KNN rods appear. Finally, platinum was sputtered on both sides as surface electrode. The rods of $\sim 40\ \mu\text{m}$ in thickness and $300\ \mu\text{m}$ in length were obtained, and the rod-to-rod distance of the present 1–3 composite sample was as small as $\sim 40\ \mu\text{m}$.²⁰¹

necessary for 1–3 composite: one is the diced piezoceramic rod arrays and the other a polymer that is filled in the grooves among the arrays.¹⁹⁷ Specific requirements for the materials include high thickness mode electromechanical coupling coefficient k_t and piezoelectric voltage coefficient g_{33} , with low mechanical quality factor Q_m and acoustic impedance Z .²⁰³

There are several methods for processing fine-scale piezoelectric ceramic/polymer composites,²⁰⁴ e.g., rod placement in a polymer matrix,²⁰⁵ dice-and-fill method,²⁰⁶ injection molding,²⁰⁷ etc.²⁰⁸ Among these approaches, “dice and fill” is established as a commercial standard due to its simplicity,¹⁹⁷ the fabrication procedure of which is shown schematically in Fig. 17.²⁰¹ During the practical implementation of KNN-based 1–3 composite, it was found that samples sintered by SPS are better in terms of dicing process than those by normal sintering due to the expected fine grain sizes and enhanced density.⁸⁷ In addition, a special treatment called “2nd dice” was found essential for good dicing quality, which means that firstly a KNN disk was diced to make a set of grooves and filled with epoxy and then the 2nd set of grooves was diced after the epoxy was cured. Fine KNN rods of $\sim 40\ \mu\text{m}$ in thickness and $300\ \mu\text{m}$ in length can be obtained, where the rod-to-rod distance is maintained at as small as $\sim 40\ \mu\text{m}$. The obtained composite sample with good dicing quality was successfully utilized as high-frequency ultrasonic transducers,¹⁹⁰ as shown in Fig. 18. Typical featured parameters include a center frequency of 29 MHz, and $-6\ \text{dB}$ bandwidths of 89.7% with the two-way IL (insertion loss) of the transducer at 29 MHz is 25.1 dB.

Similar processing procedures can also be transferred to other types of composites, e.g., ferro/magnetoelectric (ME) composites, which have drawn much attention for potential applications in sensors, transducers, and energy harvesters.^{209–214} For example, the ferromagnetic ferrites, i.e., CoFe_2O_4 (abbreviated as CFO), and KNN-based ceramics can be fabricated by two approaches to 1–3-type KNN/CFO-based ME composites. One is to utilize piezoceramic disks with arrays of square-shaped pillars backfilled with CFO and epoxy. Especially, the CFO nanoparticles were dispersed to form a CFO/ethanol suspension, and later dropped onto the

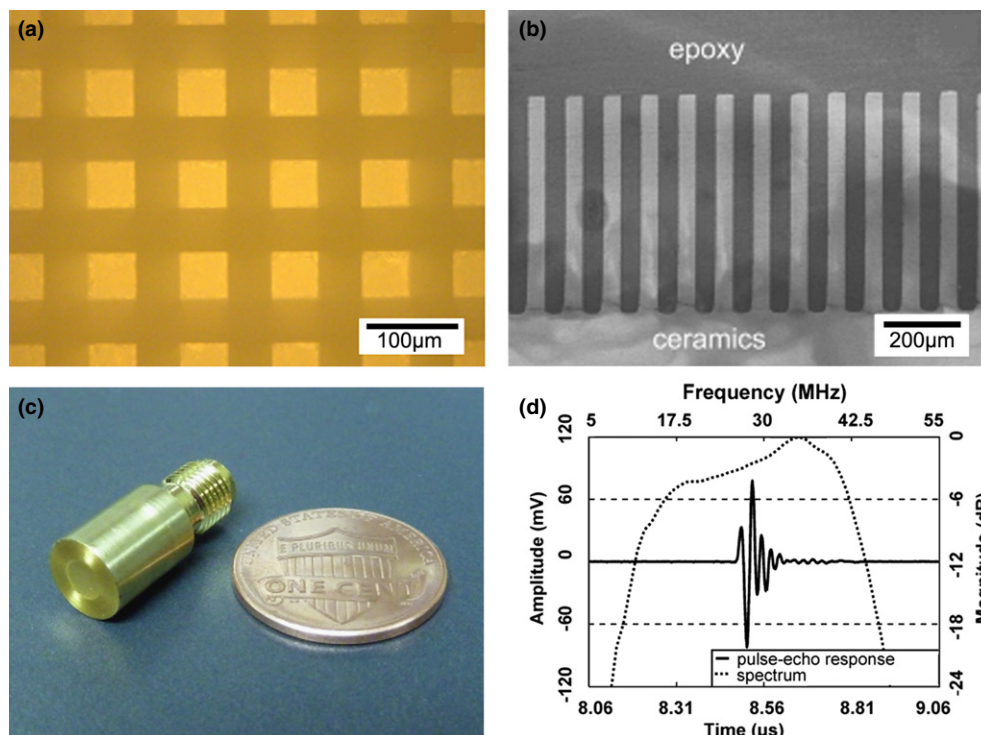


Fig. 18. (a) Optical image of the end surface, and (b) the cross-section SEM image of KNN-based rod array after dicing; (c) the photograph of 1–3 KNN/epoxy composite transducer, and (d) pulse-echo response and spectrum of the 1–3 composite for the transducer in (c).¹⁹⁰

ceramic arrays for several times to achieve highly filled 1–3 composites and the lightly filled ones at the same time as a reference.²¹⁵ This is an effective way to control the amount of CFO in the composite, and would be beneficial for the following investigation on the enhancement of ferromagnetic properties with the CFO content. The other approach is that the KNN and CFO disks with a set of parallel grooves were first fabricated, and then the two arrays with the same periodicity were carefully inserted into each other, and epoxy was also infiltrated to bond these two parts.²¹⁶ This method introduces a facile method to fabricate the 1–3 ME composite without an iterative “dice-and-fill” process.

An example of KNN/CFO-based ME composites with microscale feature size is displayed in Fig. 19. The ME voltage response α_E , defined as dV/dH , shows apparently a higher value than that of lightly filled samples as compared in Fig. 19(b), i.e., $125 \mu\text{V Oe}^{-1}$.²¹⁵ As well, the peak values of α_{E31} , α_{E32} , and α_{E33} given in Fig. 19(d) are achieved at different magnetic fields, among which α_{E31} reaches its peak of $\sim 34 \text{ mV}/(\text{cm}\cdot\text{Oe})$ at the lowest field of $\sim 790 \text{ Oe}$, while α_{E32} reaches its maximum at the highest field of $\sim 2410 \text{ Oe}$, which is attributed to the different demagnetizing behaviors in directions 1, 2, and 3 in association with the strip size.²¹⁶

(2) Single-Crystalline KNN Nanostructures

While much effort has been focused on the investigation of piezoceramics, increasing attention has been paid to the development of one-dimensional (1D) piezoelectrics, e.g., nanorod,^{198,199,217} nanowire,^{218–222} nanotube,^{223,224} etc.,²²⁵ which may be utilized in the fields of nanoelectromechanical system (NEMS), energy harvesting devices, photocatalysis, so on.^{226–229} In the case of piezoelectric materials, a wide variety of materials including ZnO, BaTiO₃, PZT, etc. have been extensively investigated. However, KNN seems to outweigh with both lead-free concept and relatively high piezoelectric performance.

Basically two methods have been utilized to fabricate the 1D piezoceramics so far: one is a template-free synthesis process, and the other is a template-assisted one. Both

molten-salt reaction and hydrothermal method belong to template-free approaches, featured by cost effectiveness, low processing temperature, and large portion of productivity; therefore they were employed in synthesizing KNN single-crystalline nanorods.²²⁶ Xu *et al.* fabricated single-crystalline niobate nanorods via molten-salt reaction, which turned out to be useful in the fabrication of niobate nanostructures.²²⁷ Afterward, the molten-salt method was transferred to making KNN nanostructures.^{230–232} It should be noted that Cheng *et al.* further showed the possibility to control the K/Na ratio of KNN nanorods, which was very important for achieving high piezoelectric output.¹⁹⁸ On the other hand, Sun *et al.* demonstrated the synthesis of $(\text{K}_{1-x}\text{Na}_x)\text{NbO}_3$ nanorods by the hydrothermal method²³³ with the growth mechanisms proposed.²³⁴ Later, Xu *et al.* reported a facile method to fabricate vertically aligned KNN nanorods, which are essentially promising for applications of energy harvesting as well as 1–3-type multifunctional composites.¹⁹⁹

Figure 20 shows a synthesis procedure of KNN nanorods via molten-salt reaction by a step-by-step process. Here, the Nb₂O₅ nanorods serve as the template for the molten-salt reaction, on top of which KNN grows to nanorods with width of 400–500 nm and length exceeding 50 μm . The morphology and structure of an individual nanorod investigated by TEM are presented in Fig. 21(a). Clear diffraction spots in the selected area electron diffraction (SAED) pattern indicate the single-crystalline nature of the obtained KNN nanorod.¹⁹⁸ The displacement of the nanorods (NR) along diameter direction, i.e., [001] direction as marked with “ $\oplus$ ” symbol in Fig. 21(a), could be measured, and used to calculate the corresponding d_{33}^* .

It is generally accepted that the piezoelectric properties of KNN materials are strongly dependent on the K/Na ratio, and reported that the KNN-based bulk materials would achieve a higher piezoelectric response when the molar contents of K and Na are accurately controlled.^{17,32,235} A similar result was also observed in a KNN single crystal. Scanning probe microscopy system was employed for measuring the dependence of displacement on voltage.²³² The displacement–voltage curve and the calculated piezoelectric response for

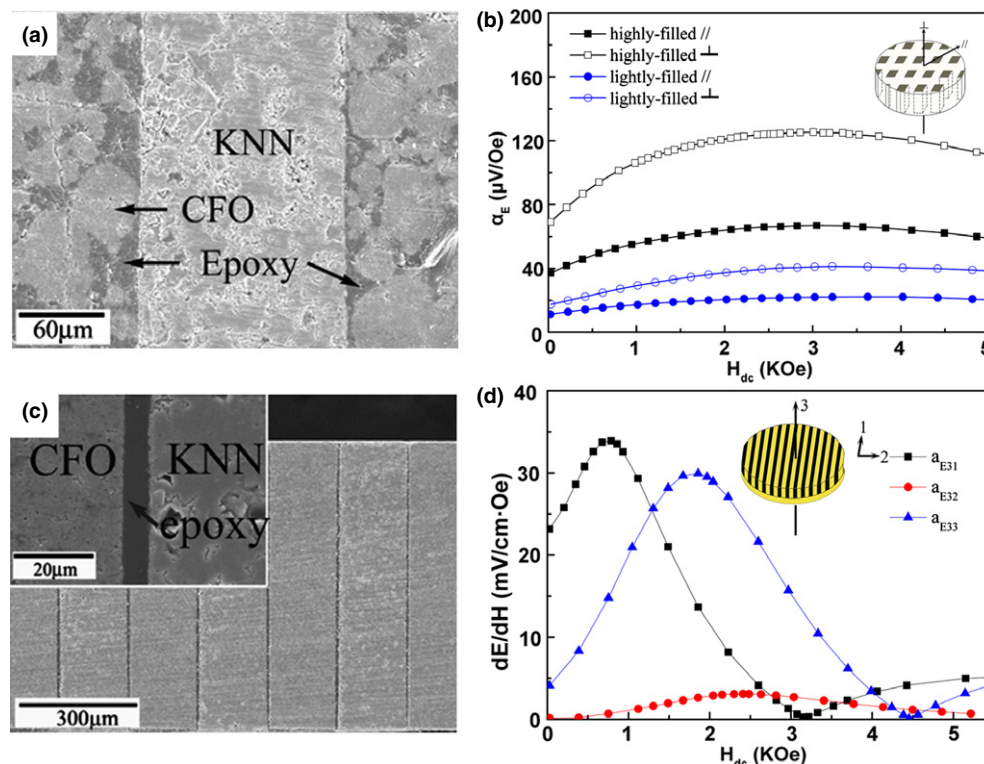


Fig. 19. (a) Cross-section morphologies of the KNN samples filled with CFO for five times, and (b) α_E values of the KNN samples filled for different times; (c) SEM images of the cross-section morphology of the strip arrays of the KNN-LT/CFO composite, and (d) the dependence of α_E values on the bias dc magnetic field H_{dc} .^{215,216}

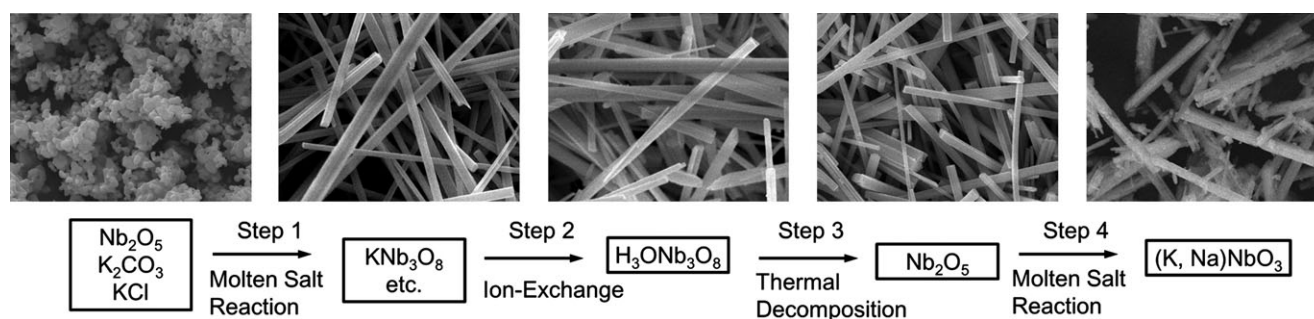


Fig. 20. Procedures for the synthesis of KNN nanorods (NR) by molten-salt reaction method. The starting materials were Nb_2O_5 , K_2CO_3 , and KCl . Niobium pentoxide first dissolved in KCl which acted as a solvent when the molten-salt reaction took place in Step 1. The nanorods with width of 400–500 nm and length exceeding 50 μm were obtained, and the rods could be assigned to $\text{K}_2\text{Nb}_8\text{O}_{21}$, KNb_3O_8 , $\text{K}_6\text{Nb}_{10.8}\text{O}_{30}$, etc., which all possessed one-dimensional morphologies. After Steps 2 and 3, the complex compound was further converted back into niobium pentoxide which inherited 1D morphology in Step 1. Finally, the Nb_2O_5 nanorods served as the template for the molten-salt reaction in Step 4, and KNN nanorods with various K/Na ratios were fabricated.¹⁹⁸

the nanorods with different K/Na ratios are shown in Figs. 21(b)–(d). A significantly high piezoelectric response with d_{33}^* up to ~ 230 pm/V is denoted.

A facile method was also employed to fabricate vertically aligned $(\text{K}, \text{Na})\text{NbO}_3$ lead-free piezoelectric nanorods.¹⁹⁹ KNN nanorods grown on KNN/STO substrate were prepared as follows. First, KNN film was grown on STO substrate by spin-coating method, and then, the obtained KNN/STO substrate was mounted on the prepared mixture which consists of potassium hydroxide (KOH), sodium hydroxide (NaOH), and niobium pentoxide (Nb_2O_5). Figure 22(a) depicts KNN nanorods grown on KNN/STO(100) substrates at 205°C for 30 h. The width of most rods was estimated at <100 nm, whereas some reached at ~ 200 –300 nm. Figure 22(b) shows the TEM observation of the KNN nanorods. A high-resolution TEM image and selected area electron diffraction (SEAD) pattern revealed that the growth direction was $<100>$. This work proposed a cost-effective method in

synthesizing Pb-free ferroelectric nanorods well aligned on substrates, which may possess higher piezoelectric response when compared with ZnO nanorod array.²³⁶ The vertically aligned KNN nanorods may have potential applications in energy-harvesting devices and multiferroic devices.

VI. Conclusions

Although relatively high piezoelectric properties were reported for pure KNN ceramics fabricated by hot pressing more than half a century ago, KNN-based lead-free piezoelectric ceramics have been refocused only in the past decade after a milestone work by Saito and his co-workers in the multidoped KNN and/or textured KNN ceramics. If compared with other lead-free candidates such as BaTiO_3 , KNN does have its advantages for applications with a higher Curie temperature and relatively lower sintering temperature as well as its better environmental compatibility. Nowadays, it

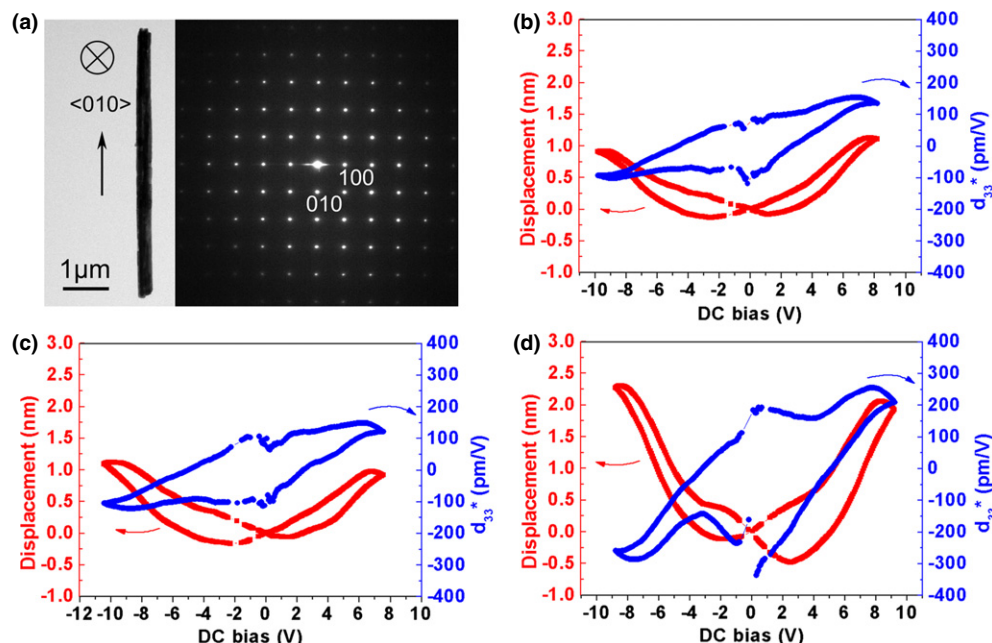


Fig. 21. (a) Diffraction patterns of TEM image for an individual KNN nanorods; and the corresponding SEAD pattern. The measurement direction of displacement in the PFM testing is indicated as “ $\oplus$ ” symbol; and displacement–voltage curve and calculated converse piezoelectric coefficient curve of KNN nanorod with the molten-salt composition of KCl/NaCl = (b)70/30, (c) 80/20, and (d) 90/10.¹⁹⁸

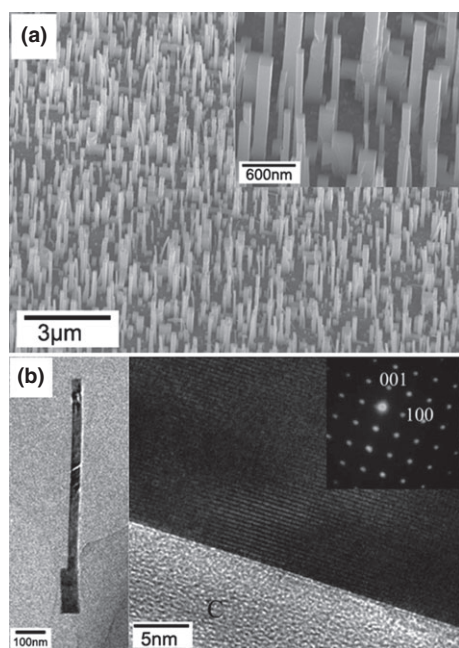


Fig. 22. (a) SEM images of KNN nanorods on (100)-oriented substrates grown for 30 h, and the substrates were tilted by 45° for observation, and (b) TEM images and the HRTEM image near the sidewall of a representative rod and its corresponding SEAD pattern.¹⁹⁹

is possible to attain high piezoelectric coefficient d_{33} over 400 pC/N even in nontextured KNN ceramics by elaborated chemical modifications, although there still remain some critical challenges for large-scale applications.

At first, creating a real MPB as in PZT system should be a breakthrough to resolve the relatively large temperature dependence of piezoelectric properties of KNN-based ceramics, which is full of scientific interest and technical importance. For this, more studies about the phase structure and phase diagram of KNN system are indispensable. Another challenge for KNN-based ceramics is that alkali elements at

A-site in KNN are easy to evaporate during sintering process. Given that the A-site chemical composition is critical to the phase control and hence property enhancement in the KNN-based ceramics, more innovative processing techniques should be continuously developed. Eventually, processing techniques should be well established to guarantee reliable mass production at an industry scale.

Finally, there have been some prototypical applications using KNN-based ceramics even in industry, which are important and encouraging for the community to push the lead-free piezoelectric research forward. We believe that lead-free piezoelectric nanomaterials will also find more applications with the advancement of micro- and nanotechnologies for energy harvesting and medical diagnostic devices. Of course, whether lead-free piezoelectric ceramics can be utilized in a large scale depends greatly on the governmental regulations on the PZT-based piezoceramics. Apart from this, more and more fundamental and practical studies for developing higher performance lead-free piezoelectric ceramics with well-balanced electrical properties and wide processing tolerance will be the key to success for the field.

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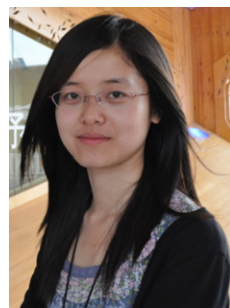


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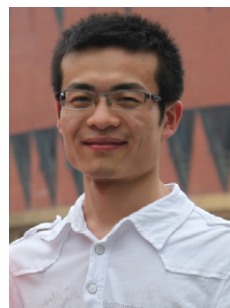
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